

Team Name: SoLowQ
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Problem Statement: The Data & AI Challenge

Solution Overview

1. Pipeline Architecture

- **Stage 1 (DuckDB):** Aggressive SQL streaming filters candidate pool based on location/role.
- **Stage 2 (NumPy/Pandas):** High-speed vectorized alignment and multi-signal scoring logic.

2. Merit-First Scoring

- **Cosine Similarity:** Semantic alignment across 12 distinct Job Description facets.
- **Verified Assessments:** Platform-certified technical skill scores scale baseline scores by [0.50, 1.20].
- **Product Exp Multiplier:** Tenure-weighted boost for product companies (+15%) vs services drag (-20%).
- **Availability Multiplier:** Notice period days mapped to a continuous suitability curve.

3. Anti-Gaming Defenses

- **Honeypot Purging:** Automatically discards fake resumes showing extreme experience gaps.
- **Sentence Co-occurrence:** Requires search & production keywords in same context window.
- **Metric Verification:** Verifies production claims against presence of digits or metrics.
- **Recruiter Action Gate:** Spam clickbaits with high views but 0 saves are demoted to Tier 4.

JD Understanding & Candidate Evaluation

Facet Type	Extracted JD Requirements
Core (70% Weight)	Retrieval Systems, Vector DB Search, Python ML Engineering, Ranking & Evaluation
Bonus (20% Weight)	LLM Fine-tuning, Learning-to-Rank, recruiting/marketplace domains, Distributed systems, Open-source ML
Penalty (-30% Weight)	Tutorial-only framework lists, Computer Vision/Speech/Robotics focus, Closed-source-only tools

Candidate Signal Evaluation

- **Verified Assessments:** Platform-certified technical skill scores (0-100) scale the baseline score. Core skills (RAG, Python) weight 1.0; ML frameworks weight 0.5.
- **Continuous Notice Curve:** Soft penalty scaling from 1.0 (<=30d notice) down to 0.80 (>=120d notice) to preserve engineering merit.
- **Product Exp Multiplier:** Product vs. consulting tenure ratio rewards SaaS/E-commerce/AI start-up experience up to +15% and drags services-only up to -20%.
- **Anti-Gaming Keywords:** Scans career histories using sentence-level syntax, requiring digits/metrics, and verifying against skill assessment scores.

Ranking Methodology

Mathematical Scoring Engine Formulas

base_fit = 0.7 * (core_score * skill_mult) + 0.2 * bonus_score - 0.3 * penalty_score + 0.05 * college_tier + 0.05 * cgpa_norm

final_score = base_fit * sqrt(availability_multiplier) * conversion_drag + location_boost

📐 1. Vector Alignment

- **Geometric Mean:** Applied over 4 core facets to avoid single-competency rescue.
- **Clipped Core:** Forces a 1e-5 floor on core facet values to prevent zero-product drops.

📈 2. Score Multipliers

- **Verified Skills:** Weights Core (1.0), Frameworks (0.5), and irrelevant (0.0). Scales base by [0.50, 1.20].
- **Product Company:** Tenure-weighted boost up to +15% and drag down to -20%.

📍 3. Gating & Location

- **Notice Curve:** Piecewise linear curve: 1.0 (<=30d) down to 0.80 floor (>=120d).
- **Location Boost:** Pune/Noida +0.015, Tier-1 +0.005. Post-merit additive boost preserves merit.

Explainability & Data Validation

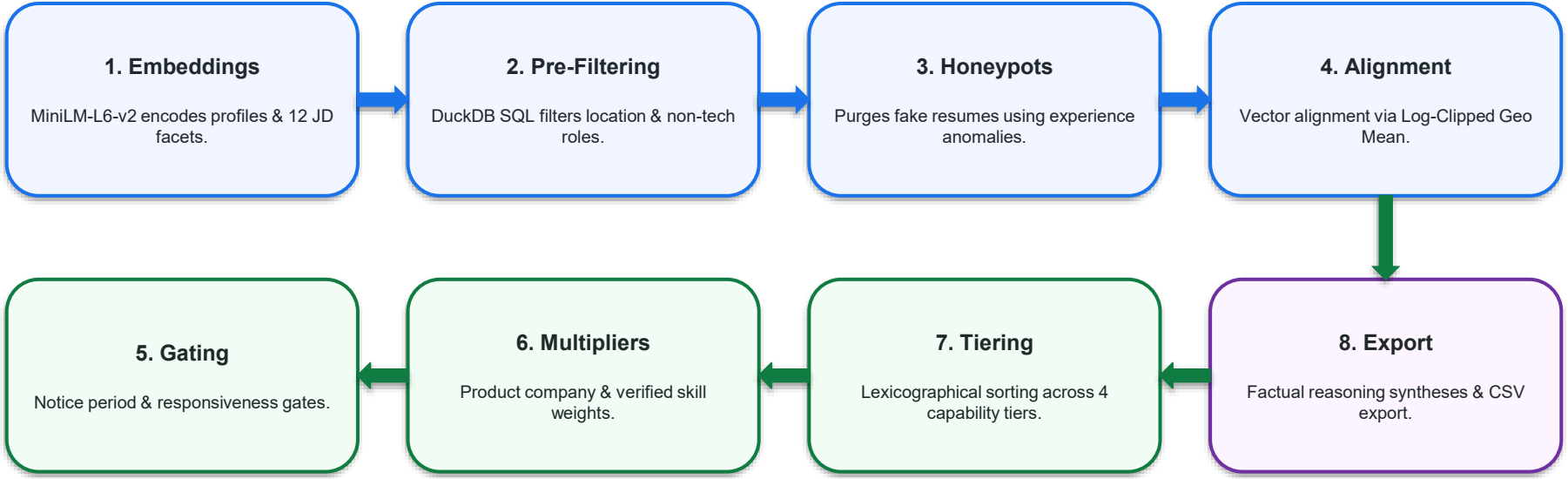
Factual Explainability Engine

- **Reasoning Generation:** Outputs plain-language recruiter summaries within 80-160 characters.
- **Strict Length Budget:** Forced compliance checks on the reasoning length to fit ATS limits.
- **Deterministic Rules:** Generates descriptions from real profile facts (years of experience, matching skill facet, location).
- **Zero Hallucination:** Eliminates generative LLM errors; utilizes structured data tables for all terminology.
- **Example Output:** "Excellent Search Engineer at Sarvam AI: 7.6y delivering embeddings retrieval, Gurgaon; clear top match for this search role."

Security & Data Integrity Defenses

- **Honeypot Purging:** Removes candidates with >12-month experience gaps or false 'expert' skills duration (<12 months).
- **Sentence Co-occurrence:** Requires search (e.g. RAG, vector search) & production (e.g. deploy, scale) keywords to co-occur in the same sentence.
- **Metric Verification:** Validates engineering claims by requiring metrics (%, million, QPS) or numbers in the sentence.
- **Anti-Stuffing Guard:** Caps technology keywords count to 8 max. Disqualifies if assessment scores are low/failed.

End-to-End Workflow



System Architecture

🔍 STAGE 1: Pre-Filtering (rank.py)

- **Input:** candidates.jsonl (100,000 candidate profiles)
- **Engine:** DuckDB in-memory SQL reader
- **Filters:** Relocation Funnel ∪ Role/Title Exclusions ∪ Consulting-only rules
- **Output:** survivors.csv (reduces candidate dataset to ~25K)

⚙️ STAGE 2: Multi-Signal Scoring (score_and_rank.py)

- **Engine:** Vectorized NumPy similarity calculations & Pandas math
- **Signals:** Geometric core similarities, verified skill multipliers, notice curves
- **Security:** Honeypot detection & anti-gaming description checking
- **Output:** submission.csv (Final ranked Top-100 candidates with reasoning)

🧩 Modular Scoring Components

🎯 Semantic Match

12 cosine similarities across core/bonus/penalty JD facets.

✅ Verified Skills

31 skill assessments scale fit by [0.50, 1.20].

⏳ Notice Period

Piecewise linear curve: 1.0 (<=30d) down to 0.80 floor.

🏢 Product Exp

Tenure-weighted SaaS/start-up ratio (+15% / -20%).

📍 Location Boost

Noida/Pune +0.015 boost preserves merit sort.

Results & Performance

Rank	Candidate ID	Score	Current Title	Company
1	CAND_0041669	1.000000	Recommendation Systems Lead	CRED
2	CAND_0064326	0.723492	Search Engineer	Sarvam AI
3	CAND_0006418	0.716009	Machine Learning Engineer	Verloop.io
4	CAND_0054394	0.692698	Recommendation Systems Lead	PharmEasy
5	CAND_0096142	0.685928	Applied ML Engineer	upGrad

Performance & Quality Validation

- **Runtime Constraints:** Stage 1 pre-filtering runs in ~45 seconds. Stage 2 scoring runs in ~90 seconds. Total runtime ~135 seconds (constraint: <5 min).
- **Memory Footprint:** Peak RAM usage stays under 4 GB on single core CPU (constraint: <16 GB).
- **Score Monotonicity:** Strict verification of score decay checks. Exactly 100 rows outputted.
- **Top 100 Quality:** 0% honeypot rate. Cleaned from keyword-stuffing and experience-inflation profiles.
- **Product Heavy:** All Top-5 matches verified as SaaS, FinTech, or AI-start-up engineers.

Technologies Used

Python 3.11

• Core pipeline scripting language with standard numerical and data library support.

DuckDB

• In-memory database engine for fast SQL streaming and parsing nested JSONL columns.

NumPy

• Vectorized matrix operations for high-speed cosine similarity score calculations.

sentence-transformers

• Encodes profiles offline with all-MiniLM-L6-v2 model (80MB, fast CPU runtime).

Pandas

• DataFrame aggregation and structural candidate data merging operations.

Apache Parquet

• Columnar parquet files provide rapid partial read access to candidate embeddings.

Submission Assets

💻 GitHub Repository

- **GitHub Repository:** Includes full project codebase and deployment instructions.
- **Link:** YODAISME/Assay

📁 Submitted Package File Structure

- **rank.py:** Stage 1 Pre-Filtering script executing role/location SQL logic.
- **score_and_rank.py:** Stage 2 Scoring script executing vectorized mathematical scoring formulas.
- **submission.csv:** The final output CSV containing the Top 100 ranked candidates.
- **precompute_embeddings.py:** Offline script generating candidate profile text embeddings.
- **submission_metadata.yaml:** YAML configuration file detailing model metadata and runtimes.

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